Khushal Bhalia

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.97
Rahul International School	Class XII (CBSE)	2020	93.60%
Gundecha Education Academy	Class X (ICSE)	2018	97.00%

WORK EXPERIENCE

Nomura Finance Ltd.

Global Markets Tech

- Built **RippleGraph** from scratch, an **AI system** that reads news and filings to predict which **250 companies** gain or lose when a supplier, customer or rival is hit by an event (fire, earthquake, big contract).
- Mapped **4,200 supplier, customer and competitor links** with **NLP and LLM extraction** from 10-K filings and Bloomberg SPLC (**88% precision**), and indexed filings and news in a **vector database** for semantic search.
- Trained a **graph neural network** that spreads each event's impact along those links, lifting next-day prediction IC from **0.015 to 0.034**; backtested on **40 past events** with point-in-time data (**0.9 Sharpe**).
- Shipped a **RAG and MCP analyst assistant** that answers "who is hurt if X shuts down?" with a ranked, source-cited list in **under 30 s** instead of **~3 hours** of manual research, used by **12 desk users**.

PROJECTS

SmartSearch - Natural Language Video Search for Dashcams | Github

Python, Pandas, SQL, Amazon Titan Embeddings, Pinecone, LLM APIs

- Built a **natural-language video search engine** over **500 real dashcam videos** for Motive's hackathon, so a query like "white truck hitting a deer on a snowy night" instantly returns every matching clip with **92% top-10 accuracy**.
- Replaced basic keyword matching (**TF-IDF**) with **Amazon Titan embeddings and Pinecone**, so the system understands meaning and not just exact words (e.g. "animal" finds "deer"), lifting accuracy from **51% to 84%**.
- Designed a **hybrid search** where an **LLM converts each query into a SQL filter** over structured data extracted from video descriptions and metadata, handling dates, places and numbers that embeddings miss, with accuracy of **91%**.
- Ran semantic ranking only on the filtered results, **cutting the search space by 85%** and returning answers in **<2 s per query**, which made the system both more accurate and faster than pure vector search.

SQLSage - Text-to-SQL Engine with Safety Guardrails | Github

Qwen3-4B, LoRA (PEFT), vLLM, sqlglot, Sentence-Transformers

- Fine-tuned a **LoRA adapter on Qwen3-4B** for schema-conditioned Text-to-SQL generation, enabling the model to generalize across unseen database schemas without retraining, trained on 2,400+ (schema, question, SQL) triples.
- Achieved **91.4% Execution Accuracy** and **84.7% Exact Match** when tested on a **Spider-style evaluation metrics**, on a held-out test set, ensuring evaluation reflected true generalization rather than memorized query patterns.
- Built a **two-stage safety guardrail** - a fine-tuned intent classifier **(98.9% recall on write-intent queries)** paired with a deterministic AST-based SQL parser - blocking unauthorized data mutation with a **99.7% combined catch rate**.
- Implemented **embedding-based schema linking** to retrieve only relevant tables for large databases, reducing irrelevant-column hallucination by ~35%.

Hidden-Rule Image Classification - Detect Objects, Then Reason | Github

Python, PyTorch, OpenCV, ResNet-18, Qwen-VL, scikit-learn, Decision Trees

- Cracked a "simple" **0/1 image classification** task on **50K train and 10K test** Blender scenes by showing it was secretly **rule-based**: a **decision tree** on **50 hand-labeled images** revealed the label depends on whether a **sphere and a cube** both appear, while end-to-end **CNNs** stalled at **~58% accuracy**.
- Built an **OpenCV pipeline** that uses color to cut every scene into **4 single-object crops**, **auto-labeled each crop's shape** with **Qwen-VL**, and trained a **multi-head ResNet-18** classifier reaching **98% accuracy** on individual objects.
- Designed a **detect-then-reason inference pipeline**: segment each test image, predict all 4 shapes, then apply the

discovered rule, reaching **97% test accuracy (+39% over the CNN baseline)** with fully **explainable predictions**.

SKILLS

- **Languages:** Java, Python, C, JavaScript, SQL, CUDA, HTML, CSS
- **Frameworks:** Pandas, NumPy, FastAPI, MCP, pytest, Docker
- **GenAI/LLMs:** LangGraph, vLLM, PyTorch, LangChain, Hugging Face Transformers, RAG
- **DevOps:** Git, GitHub, Redis, Nginx, Google Cloud Platform, Amazon Web Services
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Firebase, Redis, SQLite

SCHOLASTIC ACHIEVEMENTS

- Won **1st place** at the **Meltwater Hackathon**, across **IIT and IIIT teams**, winning a cash prize of **Rs.50,000**.
- Ranked among the **top 10 teams out of 50K+** applicants at **JPMorgan Chase's Code for Good Hackathon**, competing in the final round at its campus, and **contributing to open source** by building a solution for **Eklavya NGO** to expand access to higher education for underserved students in rural areas.
- Solved **700+ DSA questions** on Leetcode and GeekForGeeks combined along with a **1600+ rating on LeetCode**.
- Finished in top 3 at **Motive's Hackathon** by building a **hybrid fast structured-semantic video search system.**